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Glass box machine learning and corporate bond returns

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1 Big picture

- **Question.** Can corporate bond returns be predicted with machine learning while keeping the model interpretable enough to understand which variables matter and how they matter?
- **Core problem.** Machine learning models often improve forecasting performance by allowing nonlinearities and interactions, but they are usually black boxes. The paper asks whether a glass-box model can reach state-of-the-art performance without sacrificing interpretability.
- **Method.** The paper uses an Explainable Boosting Machine (EBM), an extension of a generalized additive model with selected first-order interactions. The model is additive in learned shape functions, so each feature and each interaction has a visible contribution to predicted returns.
- **Data.** The sample combines TRACE/WRDS corporate bond returns and characteristics with CRSP equity data, bond market factors, equity factors, and macroeconomic uncertainty measures. It covers July 2002 to August 2020, 218 months, 1,207 firms, 106,265 firm-month observations, and 81 predictors.
- **Benchmark result.** The EBM obtains an out-of-sample R^2 of 12.10%, close to Random Forest at 13.34% and XGBoost at 11.97%, while linear methods perform much worse.
- **Main predictors.** The most important variables are changes in financial uncertainty, changes in macroeconomic uncertainty, the Fama-French term structure factor, long-term bond reversal, and firm-level spread-related variables.
- **Main economic pattern.** Corporate bond returns react asymmetrically to uncertainty. Large increases in uncertainty predict lower returns; large declines in financial uncertainty predict higher returns, especially for risky bonds.
- **Nonlinearity.** Many predictor effects are close to zero for normal realizations and become large only in the tails. TERM has a strong positive effect mainly in its right tail; uncertainty effects are step-like rather than linear.
- **Interactions.** Interactions involving financial uncertainty and bond risk characteristics, especially spread and spread times rating, are economically important. When financial uncertainty falls, high-spread or low-grade bonds experience a disproportionately large positive expected-return contribution.
- **Heterogeneity.** Small firms, long-maturity bonds, high-spread bonds, and lower-rated bonds are more sensitive to uncertainty shocks.
- **Portfolio result.** A long-short EBM portfolio sorted on predicted returns delivers a high Sharpe ratio and transparent portfolio composition. The long leg tends to load on bonds whose characteristics the model explicitly identifies as predictive.
- **Comparison with SHAP.** EBM and SHAP-based decompositions are similar for many univariate effects, but EBM better preserves interaction structure and avoids some unstable or misleading interaction attributions.
- **Economic takeaway.** Interpretable machine learning can be a research tool, not merely a prediction tool. It reveals that corporate bond return predictability is concentrated in nonlinear uncertainty effects and in interactions between uncertainty and bond risk.

2 Data and measurement

2.1 Corporate bond return data

- The paper starts from monthly duration-adjusted corporate bond returns and characteristics provided by Kelly, Palhares, and Pruitt.
- The underlying bond data are based on TRACE through WRDS.
- The authors supplement the bond data with CRSP equity data to fill missing market capitalization information.
- Bond-level data are aggregated to the firm level using amount outstanding as the weight.
- The final return target is the next-month excess return of a firm-level corporate bond portfolio.

Interpretation: Firm-level aggregation reduces noise from multiple bonds issued by the same company while still preserving bond characteristics such as spread, duration, maturity, rating, and momentum.

2.2 Predictor universe

- The predictor set contains 81 variables: 41 firm-level variables and 40 market-level variables.
- Firm-level variables include spread, yield, duration, maturity, numerical rating, distance to default, market leverage, bond momentum, equity momentum, bond age, amount outstanding, and firm accounting measures.
- Market-level variables include Fama-French factors, momentum, corporate bond factors, liquidity factors, the VIX, inflation variables, economic policy uncertainty, Jurado-Ludvigson-Ng uncertainty measures, and other macroeconomic predictors.
- Predictors with trend behavior are divided by their monthly mean, variables censored at zero are log-transformed, and all features are standardized to mean zero and standard deviation one.

Interpretation: The paper deliberately uses a high-dimensional bond-pricing data set. The task is not only to forecast, but also to find which nonlinear relationships survive when many candidate predictors compete.

2.3 Sample split

The data are split into three time periods:

- **Training:** July 2002 to June 2010.
- **Validation:** July 2010 to June 2011.
- **Testing:** July 2011 to August 2020.

Interpretation: The long testing window is important because it includes a full out-of-sample economic cycle and the COVID period. The cost is a shorter training window, which makes the estimation noisier.

2.4 Weighting

- The main estimates weight firm observations by market capitalization.
- Equal-weighted robustness tests produce similar qualitative results, though out-of-sample performance is slightly lower.

Interpretation: Market-cap weighting reduces the influence of very small firms. The paper later studies small firms separately because they are more sensitive to uncertainty shocks.

2.5 Return moments

- The mean firm-level monthly excess return is 0.25%.
- The average firm-level monthly return standard deviation is 2.18%.
- The average monthly Sharpe ratio is 0.13, roughly 0.36 annually.
- Returns are highly time-varying: bond returns are very negative during the financial crisis and COVID onset, but high during recovery periods.

Interpretation: Corporate bond returns look modest on average but contain large cyclical swings. This makes uncertainty and credit variables natural candidates for nonlinear predictability.

2.6 Key uncertainty variables

- UNC is a macroeconomic uncertainty index from Jurado, Ludvigson, and Ng.
- $UNCf$ is the financial uncertainty index.
- $UNCr$ is the real-economy uncertainty index.
- The paper emphasizes first differences, especially $\Delta UNCf$ and ΔUNC .

Interpretation: Changes in uncertainty matter more than levels. A rise in uncertainty predicts lower next-month corporate bond returns, while a decline in financial uncertainty predicts higher returns for riskier bonds.

2.7 Important bond risk characteristics

- **spread:** the firm-level bond yield spread.
- **sprxrtg:** spread multiplied by numerical rating, where higher ratings codes mean worse credit quality.
- **sprd2d:** spread-to-distance-to-default.
- **momspr:** spread momentum.
- **maturity/duration:** bond maturity and duration measures.

Interpretation: The strongest interactions occur when aggregate uncertainty variables meet firm-level bond risk variables. This is economically intuitive: uncertainty shocks matter most for bonds that are already risky or difficult to value.

3 Models and empirical specifications

3.1 General prediction equation

The starting point is the relationship between next-month excess bond returns and information at time t :

$$r_{l,t+1} = g(\mathbf{z}_{l,t}, \mathbf{z}_t) + e_{l,t+1}, \quad (1)$$

where $\mathbf{z}_{l,t}$ is a vector of bond and firm characteristics, and $\mathbf{z}_t$ is a vector of market-level predictors.

Interpretation: The object of interest is a high-dimensional function $g(\cdot)$ mapping bond, firm, market, and macro variables into expected next-month excess returns.

3.2 EBM additive structure

The EBM estimates a generalized additive model with selected pairwise interactions:

$$r_{l,t+1} = \beta + \sum_{i=1}^N f_i(x_{i,l,t}) + \sum_{i>j} f_{ij}(x_{i,l,t}, x_{j,l,t}) + e_{l,t+1}. \quad (2)$$

- $f_i(\cdot)$ is a univariate shape function for predictor i .
- $f_{ij}(\cdot, \cdot)$ is a shape function for an interaction between predictors i and j .
- Shape functions are step functions estimated with shallow decision trees, bagging, and boosting.

Interpretation: The model is flexible because each effect can be nonlinear. It is interpretable because the forecast is a sum of visible components.

3.3 Model evaluation

The paper evaluates models by the out-of-sample R^2 :

$$R_{\mathcal{T}}^2 = 1 - \frac{\sum_{(l,t) \in \mathcal{T}} e_{l,t}^2}{\sum_{(l,t) \in \mathcal{T}} r_{l,t+1}^2}. \quad (3)$$

Interpretation: The denominator does not demean returns, so the benchmark forecast is zero excess return. This prevents the model from looking good merely because it beats a weak mean benchmark.

3.4 Mean absolute scores

The native EBM importance measure is the weighted mean absolute contribution of a shape function:

$$S(i) = \sum_{(l,t) \in \mathcal{T}_1} w_{l,t} |f_i(x_{i,l,t})|. \quad (4)$$

For interactions,

$$S(i, j) = \sum_{(l,t) \in \mathcal{T}_1} w_{l,t} |f_{ij}(x_{i,l,t}, x_{j,l,t})|. \quad (5)$$

Interpretation: $S(i)$ is analogous to the absolute size of a standardized regression coefficient, but it can measure nonlinear effects and interactions.

3.5 Importance from removing variables

The paper also removes a fitted term and recomputes R^2 . For a single univariate function, the change is

$$\Delta R_i^2 = R_{-i}^2 - R^2. \quad (6)$$

The total importance of variable i removes both its univariate function and all interactions involving i :

$$\Delta \tilde{R}_i^2 = \tilde{R}_{-i}^2 - R^2. \quad (7)$$

Interpretation: This asks how much predictive fit is lost if a predictor is unavailable. For $\Delta UNCf$, the total loss is larger than the univariate loss because interactions involving financial uncertainty are important.

3.6 Portfolio construction

For each month t , firms are sorted into deciles using their EBM predicted return $\hat{r}_{l,t+1}$. The standard EBM portfolio buys the top decile and shorts the bottom decile. The EBM-pos version invests in the risk-free asset when the predicted return of a decile portfolio is negative.

Interpretation: The portfolio exercise asks whether EBM predictability is economically useful and whether the model's transparent shape functions explain the resulting long and short positions.

3.7 SHAP comparison

For black-box models, the paper computes SHAP decompositions:

$$\hat{r}_{l,t+1} = \phi_0 + \sum_{i=1}^N \phi_{i,l,t} + \sum_{i>j} \phi_{ij,l,t}. \quad (8)$$

Interpretation: SHAP produces post-hoc explanations for Random Forest and XGBoost. EBM instead has built-in global shape functions. The comparison asks whether post-hoc explanations recover similar economics.

3.8 Simulation design

The simulation creates nonlinear univariate effects and interactions, then compares EBM, Random Forest SHAP, and XGBoost SHAP in recovering the true effects. Model fit is measured by R^2 , variable selection, and RMSE relative to the data-generating process.

Interpretation: The simulation is a controlled laboratory. It checks whether the attribution methods recover the true nonlinear interaction structure, not just whether they forecast well.

4 Identification and empirical design

4.1 What the paper identifies

- The paper identifies out-of-sample predictive relationships between corporate bond returns and a large predictor set.
- It does not claim a causal effect of uncertainty on bond returns in the experimental sense.
- It does show that a small set of uncertainty and spread-related variables contains stable predictive information.

Interpretation: This is a predictive asset pricing paper. The credible design feature is out-of-sample evaluation, not randomized variation.

4.2 Why the out-of-sample split matters

- Hyperparameters are chosen in the validation sample and performance is evaluated in the later test sample.
- Random Forest and XGBoost have much higher training R^2 than EBM but similar or slightly worse test performance, indicating overfitting risk.
- EBM has lower training fit but remains competitive out of sample.

Interpretation: The results support the paper's main claim: interpretability need not come with a large performance penalty.

4.3 Why uncertainty is not just generic macro overfitting

- The paper replaces uncertainty variables with PCA macro factors from the FRED-MD panel.
- When only PCA macro factors are included, out-of-sample performance falls sharply.
- When the original uncertainty variables are added back, performance improves again.
- Ridge-style macro factors also fail to replicate the uncertainty variables' predictive content.

Interpretation: The uncertainty measures are not merely proxies for a generic macro data set. They appear to summarize a distinct state variable for corporate bond pricing.

4.4 Why the shape functions matter

- A linear model could say that uncertainty has a coefficient.
- The EBM shows that the effect is mostly inactive in the center of the distribution and large in the tails.
- The EBM also shows whether an effect is symmetric or asymmetric.

Interpretation: Economic interpretation changes when the relation is nonlinear. The paper's main economics are tail effects, not average linear slopes.

4.5 Why interactions matter

- Interaction terms account for 89% of the total EBM score.
- Financial uncertainty by itself matters, but its interaction with spread and rating is central.
- Falling financial uncertainty generates high expected returns primarily for risky bonds.

Interpretation: The model predicts states of the world and types of bonds jointly. Risky bonds are not always attractive; they are attractive when uncertainty falls.

4.6 What the paper cannot fully prove

- Machine learning forecasts do not automatically imply risk premia rather than mispricing or microstructure effects.
- The EBM is interpretable but still estimated from finite-sample data, so shape functions can be noisy in sparse tails.
- The paper uses monthly firm-level bond returns; some bond-level trading frictions and transaction costs are outside the main prediction model.
- The model is transparent relative to black-box alternatives, but it remains a statistical forecasting model rather than a fully structural equilibrium model.

Interpretation: The paper’s strongest contribution is not a causal theory of uncertainty. It is the demonstration that interpretable nonlinear functions can match black-box performance and reveal economically meaningful return patterns.

5 Tables and figures

5.1 Figure 1 and Tables 1–3: EBM mechanics and baseline model performance

Read:

- Figure 1 illustrates the EBM algorithm: the model cycles through features, fits shallow trees, updates residuals, and accumulates feature-specific shape functions.
- Table 1 shows that average firm-level excess returns are positive but noisy: mean return is 0.25% per month and mean return volatility is 2.18%.
- Table 2 shows that OLS has negative out-of-sample R^2 , LASSO improves to 7.99%, and nonlinear ML models reach about 12%–13%.
- EBM achieves 12.10% out-of-sample R^2 , close to Random Forest and XGBoost while being interpretable.
- Table 3 shows that EBM forecast errors are closest to zero on average among the models.

Economic message: The model comparison establishes the paper’s central performance claim: a glass-box model can match black-box methods in corporate bond return prediction.

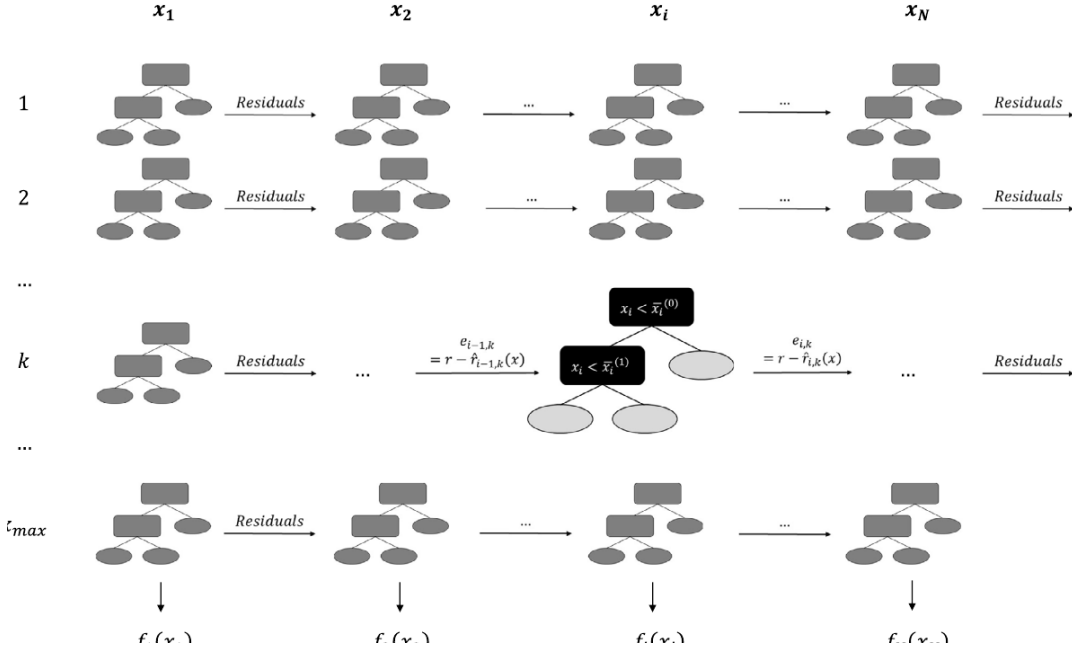


Figure 1: EBM algorithm

Table 1

Descriptive statistics of excess returns.

	Mean	S.D.	5% pct.	95% pct.
Mean of $r_{i,t}$ by firm	0.25	0.22	-0.00	0.64
S.D. of $r_{i,t}$ by firm	2.18	1.08	0.72	4.21
SR of $r_{i,t}$ by firm	0.13	0.10	-0.00	0.31

Note: This table reports means, standard deviations, and the 5% and 95% percentiles of the distribution of moments of excess bond returns. The rows show the distributions of the means, standard deviations, and Sharpe ratios of excess returns by firm. Firms with less than 24 observations in the sample are excluded. The sample is from July

Table 2

Comparison of R^2 's.

	OLS	LASSO	R-Forst	XGBoost	EBM
R^2_{train}	31.15%	17.51%	46.93%	42.52%	24.79%
R^2_{out}	-0.27%	15.94%	15.57%	21.80%	18.54%
R^2_{test}	-2.34%	7.99%	13.34%	11.97%	12.10%

Table 1: Descriptive statistics of excess returns

Table 2: Model comparison

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Table 3

Model errors.

	OLS	LASSO	R-Forst	XGBoost	EBM
Panel A: Descriptive Statistics					
Mean	-0.62	0.03	0.08	0.11	0.02
S.D.	2.15	2.12	2.06	2.07	2.08
Skewness	-1.64	-0.89	-0.60	-0.77	-0.96
Kurtosis	18.19	21.76	20.12	20.41	21.82
Panel B: Correlations					
OLS	1.00	0.91	0.88	0.88	0.90
LASSO	0.91	1.00	0.98	0.96	0.99

Table 3: Model errors

5.2 Figures 2–3: predictor importance and parsimonious predictability

Read:

- Figure 2 shows that $\Delta UNCf$, ΔUNC , TERM, and LTREVB are the most important predictors.
- The total R^2 loss from removing $\Delta UNCf$ and its interactions is about 4.36 percentage points.
- TERM has the largest univariate R^2 effect, but $\Delta UNCf$ has the largest total effect because it participates in important interactions.
- Figure 3 shows that $\Delta UNCf$ alone obtains 4.51% out-of-sample R^2 .
- Adding TERM, ΔUNC , and spread raises out-of-sample R^2 to 10.01%.
- The top ten variables produce 11.78%, very close to the full 81-variable model's 12.10%.

Economic message: Most corporate bond return predictability is concentrated in a small number of uncertainty, term structure, and spread variables. High dimensionality helps, but only a few variables carry most of the signal.

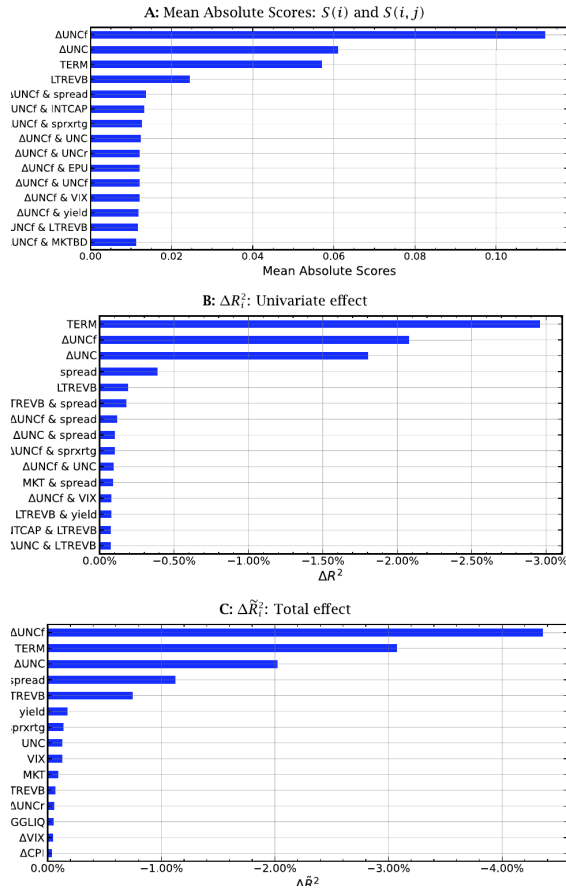


Figure 2: Importance of predictors

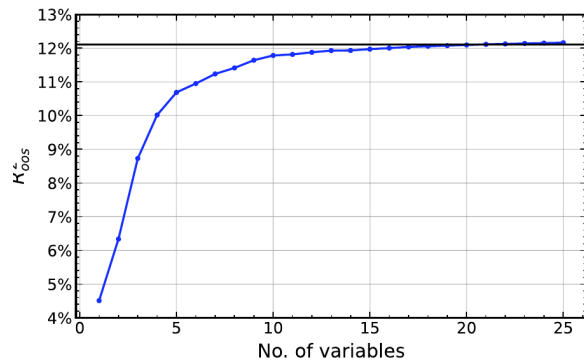


Figure 3: Number of predictors and out-of-sample R^2

5.3 Figures 4–7: nonlinear uncertainty and spread effects

Read:

- Figure 4 shows tail-driven univariate effects. Large declines in financial uncertainty predict return premia of roughly 64 bps, while large increases in financial uncertainty predict penalties of roughly 65 bps.
- The TERM effect is almost zero over most of the distribution but jumps to roughly 100 bps in the right tail.
- Spread has a monotone positive effect, but the effect becomes meaningful only for high-spread bonds.
- Figures 5 and 6 show that $\Delta UNCf$ interacts strongly with spread and spread times rating.
- When financial uncertainty falls, high-spread or low-grade bonds earn an additional positive contribution; when uncertainty rises, the penalty is smaller and more broadly spread across bonds.
- Figure 7 shows that observed and predicted returns line up with this interaction pattern: high-risk bonds do well when financial uncertainty falls and poorly when it rises.

Economic message: The core pricing result is state dependent. Risky bonds are attractive mainly in states where financial uncertainty declines, consistent with price recovery after uncertainty subsides.

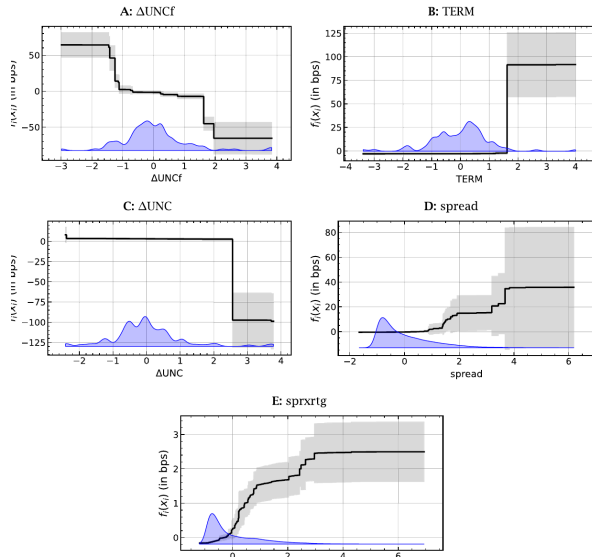


Figure 4: Univariate shape functions

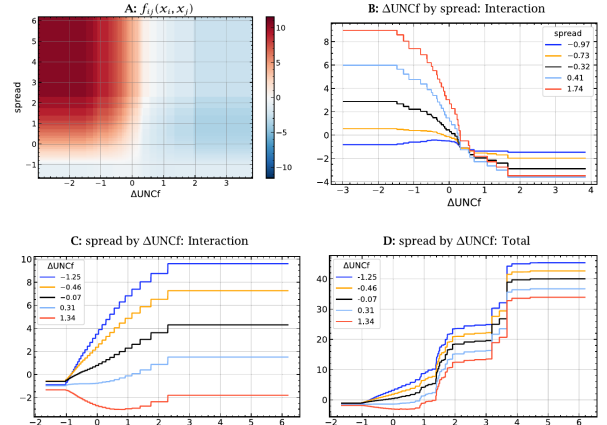


Figure 5: $\Delta UNCf$ and spread interaction

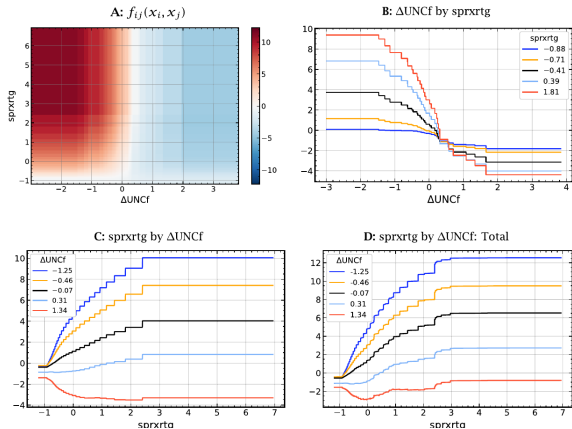


Figure 6: $\Delta UNCf$ and spread-times-rating interaction

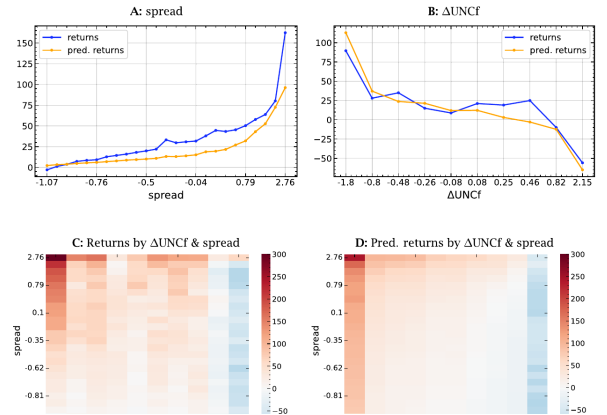


Figure 7: Observed and predicted returns by uncertainty and spread

5.4 Figures 8–10 and Table 4: uncertainty levels, firm heterogeneity, and macro robustness

Read:

- Figure 8 shows that changes in financial uncertainty are more predictive than levels. The correlation between $\Delta UNCf$ and next-month bond returns is strongly negative.
- Figure 9 shows that small firms, high-spread bonds, worse-rated bonds, high spread-to-distance-to-default bonds, high spread-times-rating bonds, and long-maturity bonds are more sensitive to financial uncertainty.
- Table 4 shows that PCA macro factors by themselves do not replicate the predictive power of uncertainty variables.
- Figure 10 confirms that even after adding macro PCA factors, the importance ranking remains dominated by $\Delta UNCf$, ΔUNC , and TERM.

Economic message: Uncertainty is not a generic macro proxy. It is a distinct predictor whose effect is strongest exactly where credit risk is more pronounced.

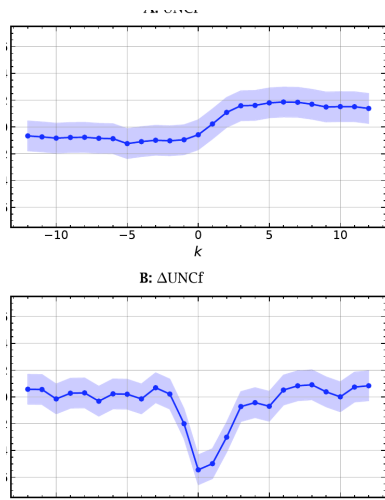


Figure 8: Cross-correlation of uncertainty and returns

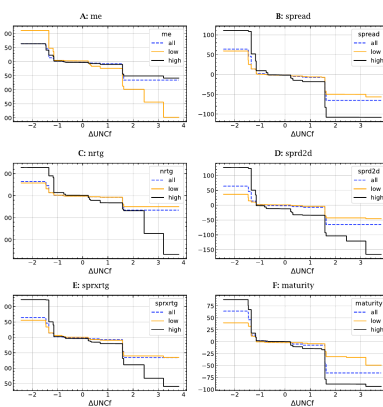


Figure 9: $\Delta UNCf$ effects by subset

Table 4
Comparison of R^2 's

	OLS	LASSO	R-forest	XGboost	EBM
Panel A: Including PCA macro factors					
R^2_{train}	28.92%	18.54%	46.67%	20.41%	26.38%
R^2_{val}	8.28%	9.46%	6.57%	9.39%	8.30%
R^2_{test}	-15.83%	3.32%	6.35%	4.68%	6.38%
Panel B: Including PCA macro factors and uncertainty variables					
R^2_{train}	32.05%	17.62%	36.33%	28.16%	25.47%
R^2_{val}	-2.75%	15.65%	13.69%	10.13%	16.31%
R^2_{test}	-9.63%	7.96%	13.25%	11.36%	11.78%
Panel C: Including ridge macro factor					
R^2_{train}	27.44%	18.30%	53.44%	19.87%	26.04%
R^2_{val}	0.17%	9.59%	8.09%	12.24%	12.55%
R^2_{test}	-11.46%	3.45%	4.61%	5.72%	6.37%

This table shows R^2 's of OLS, LASSO, Random Forest, XGboost, and EBM estimations in the training, validation, and test samples when refitting models on different variations of our data set. Panels A

Table 4: Macro PCA robustness

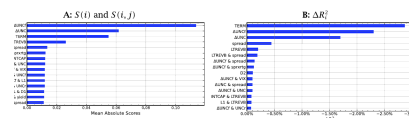


Figure 10: Importance with PCA factors

5.5 Tables 5–6 and Figure 11: EBM portfolio performance

Read:

- Table 5 reports decile portfolios sorted on EBM predicted returns.
- The value-weighted EBM long-short portfolio has mean monthly return 0.67%, volatility 1.75%, and monthly Sharpe ratio 0.38.
- The equal-weighted version has a monthly Sharpe ratio of 0.40.

- Figure 11 shows that the EBM portfolio outperforms benchmark characteristic and factor portfolios over the test period.
- Table 6 shows that the best characteristic portfolios are related to spread momentum, spread, and spread times rating; the best factor portfolios include MKT B, PEADB, and TERM.

Economic message: The forecasting model is economically useful: predicted-return sorts generate strong long-short performance, and the long leg can be explained using the model’s visible rules.

Table 5
Returns of EBM portfolios.

	1	2	3	4	5	6	7	8	9	10	10-1
Mean	-0.06	0.04	0.09	0.12	0.16	0.21	0.21	0.28	0.37	0.61	0.67
S.D.	1.28	1.22	1.12	1.25	1.37	1.50	1.71	1.92	2.09	2.60	1.75
SR	-0.05	0.03	0.08	0.10	0.12	0.14	0.12	0.14	0.18	0.23	0.38
Panel B: EBM-pos VW											
Mean	0.04	0.10	0.15	0.20	0.25	0.34	0.37	0.44	0.50	0.68	0.64
S.D.	0.40	0.55	0.62	0.73	0.80	0.96	1.10	1.20	1.46	1.96	1.72
SR	0.10	0.18	0.25	0.28	0.31	0.35	0.34	0.37	0.35	0.35	0.37
Panel C: EBM EW											
Mean	-0.08	0.02	0.09	0.11	0.15	0.18	0.22	0.30	0.38	0.65	0.73
S.D.	1.27	1.19	1.12	1.22	1.38	1.52	1.70	1.88	1.99	2.67	1.81
SR	-0.07	0.02	0.08	0.09	0.11	0.12	0.13	0.16	0.19	0.24	0.40
Panel D: EBM-pos EW											
Mean	0.03	0.09	0.15	0.19	0.25	0.32	0.38	0.46	0.52	0.74	0.71
S.D.	0.35	0.46	0.55	0.67	0.75	0.91	1.05	1.18	1.39	1.97	1.77
SR	0.09	0.20	0.27	0.29	0.34	0.36	0.36	0.39	0.37	0.38	0.40

Note: This table compares monthly means, standard deviations, and Sharpe ratios for ten value-weighted portfolios constructed based on predicted return deciles in the testing sample. The last column shows the results of the 10-1 portfolio, which is long in portfolio 10 and short in portfolio 1. EBM-pos equals the combined portfolio, and VW-pos equals combined portfolio by bond market value. EBM-ew is the return-weight

Table 5: Returns of EBM portfolios

Table 6

Returns of characteristics-sorted portfolios and factors.

Panel A: 5-1 characteristic portfolios										
	momspr	spread	sprxrtg	nime	yield	spr2d	coupon	voleq	ret	skew
Mean	0.29	0.45	0.39	0.10	0.42	0.26	0.24	0.21	0.22	0.13
S.D.	0.98	1.61	1.44	0.39	1.77	1.23	1.20	1.15	1.52	0.98
SR	0.30	0.28	0.27	0.26	0.24	0.21	0.20	0.18	0.15	0.13
Panel B: Factors										
	MKTB	PEADB	TERM	CRY	VAL	DRF	HMLB	RMW	LTREVB	SMB
Mean	0.44	0.12	0.62	0.52	0.37	0.40	0.15	0.14	-0.04	-0.24
S.D.	1.58	0.52	3.02	2.58	1.97	2.34	1.62	1.55	0.92	2.44
SR	0.28	0.24	0.20	0.20	0.19	0.17	0.09	0.09	-0.05	-0.10

Note: This table reports monthly means, standard deviations, and Sharpe ratios of characteristics-sorted portfolios and factors. We sort characteristics-sorted portfolios and factors by their Sharpe ratios in the combined training and validation samples. We then report statistics of the ten characteristics-sorted portfolios (Panel A) and factors (Panel B) in the testing sample.

Table 6: Returns of characteristic-sorted portfolios and factors

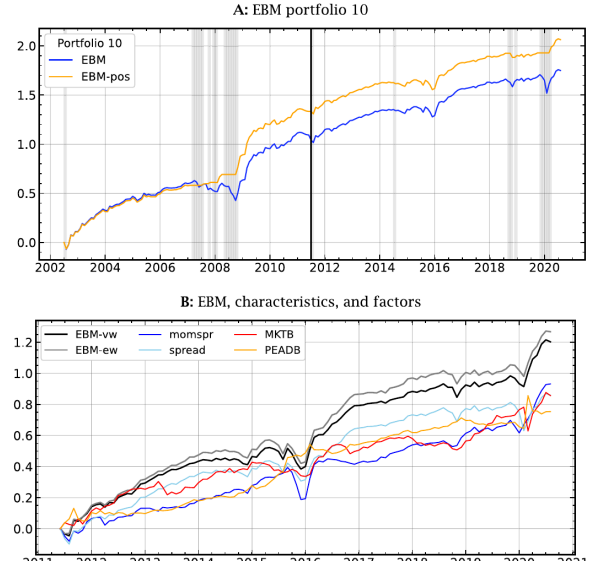


Figure 11: Cumulative returns

5.6 Figures 12–13: what the EBM portfolios buy and short

Read:

- Figure 12 shows that EBM portfolio deciles are monotonic in important model variables such as spread, while variables with low model importance are not systematically sorted.
- Portfolio 10 consistently contains bonds with high spreads, while portfolio 1 contains low-spread bonds.
- Figure 13 shows that the long leg’s spread and spread-times-rating move negatively with $\Delta UNCf$; the model buys riskier bonds when financial uncertainty falls.
- The short leg has the opposite relation.

Economic message: The EBM portfolio is not a black-box trade. Its composition follows transparent economic rules: it leans into credit risk when uncertainty falls and avoids it when uncertainty rises.

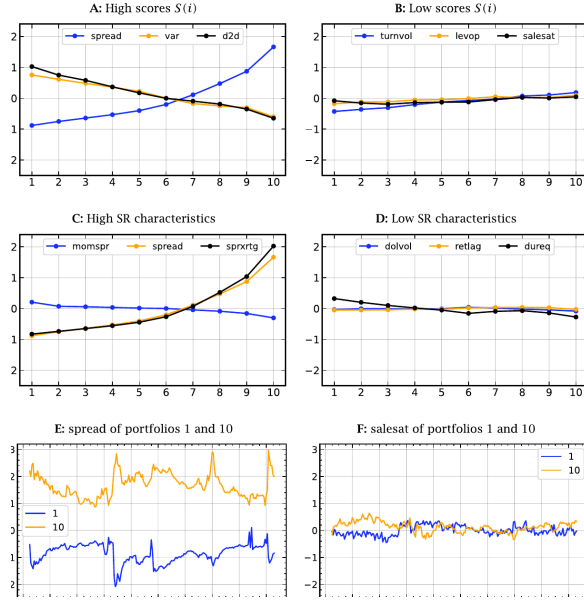


Figure 12: Characteristics of EBM portfolios

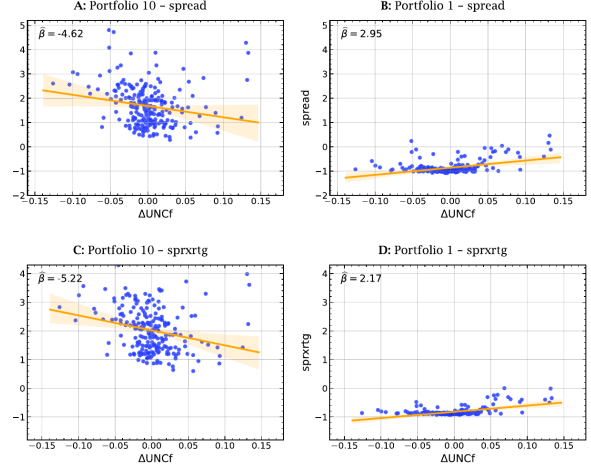


Figure 13: $\Delta UNCf$ and EBM portfolio characteristics

5.7 Tables 7–8 and Figure 14: EBM versus SHAP explanations

Read:

- Table 7 lists the top ten variables or effects in EBM, Random Forest, and XGBoost. EBM's top variables are dominated by $\Delta UNCf$, ΔUNC , TERM, LTREVB, and interactions involving $\Delta UNCf$.
- Table 8 shows that univariate EBM effects and SHAP effects are highly correlated, but interaction effects have much weaker agreement.
- Figure 14 illustrates why: SHAP values can be more local, noisy, and heterogeneous, while EBM produces stable global shape functions.
- The interaction between $\Delta UNCf$ and spread is clear and monotone in EBM but less transparent under SHAP decompositions.

Economic message: Post-hoc explainability is useful, but it is not the same as an inherently interpretable model. The differences are most important for interaction effects, which are central to bond return predictability.

Table 7

Top 10 variables.

	EBM	Random Forest	XGBoost
1	$\Delta UNCf$	$\Delta UNCf$	$\Delta UNCf$
2	ΔUNC	levmkt	spread
3	TERM	ΔUNC	
4	LTREVB	TERM	ret
5	$\Delta UNCf$, spread	UNC	ΔUNC
6	$\Delta UNCf$, INTCAP	spread	$\Delta UNCf$, spread
7	$\Delta UNCf$, spravg	$\Delta UNCf$	$\Delta UNCf$, $UNCf$
8	$\Delta UNCf$, UNC	LTREVB	VIX
9	$\Delta UNCf$, $UNCf$	EPUT	$\Delta UNCf$, UNC
10	$\Delta UNCf$, EPU	VIX	MOMB

Notes: This table presents the top 10 EBM, Random Forest, and XGBoost effects.

Table 7: Top variables

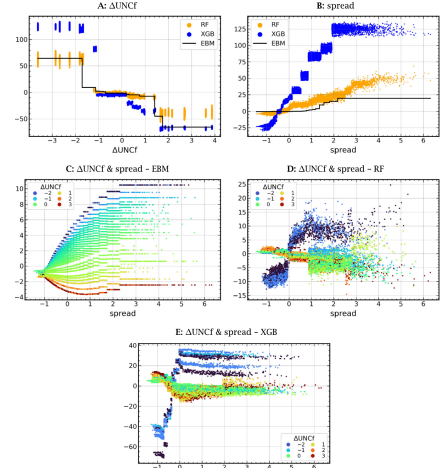
Table 8

Top 10 effects — Comparison of EBM and SHAP.

	EBM vs. RF	EBM vs. XGB	RF vs. XGB
Panel A: Correlations			
$\Delta UNCf$	0.9654	0.8976	0.9046
ΔUNC	0.9953	0.9992	0.9954
TERM	0.9812	0.9726	0.9707
LTREVB	0.9708	0.6651	0.7488
$\Delta UNCf$ & spread	0.3566	0.4147	0.5605
$\Delta UNCf$ & INTCAP	0.4930	0.2124	0.4994
$\Delta UNCf$ & spravg	0.3573	—	—
$\Delta UNCf$ & UNC	0.4909	0.5184	0.7100
$\Delta UNCf$ & $UNCf$	0.3695	-0.0334	-0.2122
$\Delta UNCf$ & EPU	0.0830	-0.3423	0.1679
Panel B: RMSE			
$\Delta UNCf$	0.146	0.397	0.439
ΔUNC	0.052	0.219	0.207
TERM	0.145	0.472	0.388
LTREVB	0.190	0.869	0.894
$\Delta UNCf$, spread	0.575	0.747	0.722
$\Delta UNCf$, INTCAP	0.512	0.629	0.503
$\Delta UNCf$, spravg	0.686	1.046	1.003
$\Delta UNCf$, UNC	0.518	0.889	0.829
$\Delta UNCf$, $UNCf$	0.576	0.950	0.991
$\Delta UNCf$, EPU	0.767	0.833	0.821

Note: This table presents correlations (Panel A) and root mean squared errors (Panel B), defined in (10), between EBM outputs $f(\mathbf{x}_{i,j})$ and SHAP decompositions $\phi_{i,j}$ for the

Table 8: EBM versus SHAP

Figure 14: SHAP values for $\Delta UNCf$ and spread

5.8 Tables 9–11 and Figure 15: decomposition and simulation evidence

Read:

- Table 9 shows that long-short portfolios sorted on EBM effects produce strong Sharpe ratios for both univariate effects and interactions.
- Table 10 decomposes the total score: interactions account for 89% of the EBM score, compared with 66% for Random Forest SHAP and 58% for XGBoost SHAP.
- Mixed firm-market interactions account for a large share of EBM importance.
- Table 11 shows the simulation results: EBM has the highest test R^2 among the three methods in the simulation and the lowest RMSE for every listed effect.
- Figure 15 shows that EBM recovers nonlinear interaction effects more accurately than RF or XGBoost SHAP values.

Economic message: Interaction structure is not a side issue. It is the dominant source of EBM explanatory power, and simulations show that the EBM architecture recovers interactions more faithfully.

Table 9
Returns of long-short portfolios sorted on EBM and SHAP effects.

Panel A: EBM — univariate										
	spread	sprxrtg	spread	yield	age	voleq	skew	levmkt	retlag	ret
Mean	0.33	0.51	0.61	0.39	0.10	0.18	0.15	0.09	0.16	0.17
S.D.	1.03	1.71	2.06	1.45	0.44	1.24	1.03	0.66	1.64	1.92
SR	0.32	0.30	0.30	0.27	0.23	0.15	0.14	0.13	0.10	0.09
Panel B: Random Forest — univariate										
	me	amtout	nrtg	coupon	dolvol	beme	maturity	offramt	sales	duration
Mean	0.37	0.24	0.33	0.10	0.07	0.04	0.06	0.07	0.03	0.03
S.D.	1.00	0.66	1.07	0.75	0.59	0.51	0.77	1.09	0.74	0.66
SR	0.37	0.35	0.31	0.13	0.12	0.08	0.08	0.06	0.05	0.04
Panel C: XGBoost — univariate										
	assets	spread	momspr	nrtg	ret	offramt	atme	sprxrtg	voleq	maturity
Mean	0.15	0.57	0.21	0.20	0.30	0.24	0.15	0.05	0.07	0.04
S.D.	0.50	1.93	1.15	1.13	1.65	1.35	0.91	0.47	1.08	0.68
SR	0.30	0.30	0.19	0.18	0.18	0.18	0.17	0.11	0.07	0.06
Panel D: EBM — interactions										
	profitbl,	ret,	maturity,	duration,	ret,	nime,	dureq,	ret,	ret,	mom,
Mean	0.37	0.49	0.41	0.36	0.39	0.35	0.31	0.30	0.30	0.22
S.D.	0.82	1.09	1.01	0.92	1.36	1.27	1.12	1.29	1.45	1.71
SR	0.46	0.45	0.41	0.40	0.28	0.28	0.27	0.23	0.21	0.13
Panel E: Random Forest — interactions										
	totaldebt,	dolvol,	duration,	mom,	momspr,	levop,	assets,	levmkt,	ret,	debtebtida,
Mean	0.25	0.07	0.09	0.11	0.06	0.04	0.04	0.02	0.02	0.01
S.D.	0.85	0.62	0.80	1.22	0.79	0.57	0.62	0.51	0.94	0.86
SR	0.30	0.11	0.11	0.09	0.07	0.07	0.06	0.04	0.02	0.01
Panel F: XGBoost — interactions										
	beme,	spread,	atme,	d2d,	atbe,	nime,	nime,	offramt,	amtout,	amtout,
Mean	0.31	0.38	0.37	0.22	0.18	0.11	0.09	0.16	0.07	0.07
S.D.	0.77	0.96	1.07	1.01	0.81	0.58	0.48	0.97	0.54	0.67
SR	0.31	0.38	0.37	0.22	0.18	0.11	0.09	0.16	0.07	0.07

Table 9: Long-short portfolios sorted on EBM and SHAP effects

Table 10
Decompositions of effects.

Variables	EBM	RF	XGB
Univariate	0.11	0.34	0.42
Interactions	0.89	0.66	0.58
Firm	0.17	0.17	0.18
Mkt	0.36	0.53	0.54
Firm/mkt	0.47	0.30	0.28
Univariate-firm	0.01	0.09	0.13
Univariate-mkt	0.10	0.25	0.29
Interaction-firm	0.16	0.08	0.05
Interaction-mkt	0.26	0.29	0.25
Interaction-firm/mkt	0.47	0.30	0.28

Note: This table shows the decompositions of the total score of the EBM model S at the total SHAP values ϕ of the RF and XGB models.

Table 10: Decompositions of effects

Table 11
Simulation results.

	EBM	RF	XGB
Panel A: R^2			
Train	14.25	12.88	15.90
Valid	10.69	9.42	10.28
Test	9.96	8.49	8.65
Panel B: Term selection			
z_1	1.00	1.00	1.00
y	0.90	0.98	0.97
(z_1, y)	1.00	1.00	0.99
(z_2, y)	1.00	0.99	0.97
Panel C: RMSE			
z_1	1.47	1.56	1.50
y	0.18	0.30	0.25
(z_1, y)	0.84	1.05	1.13
(z_2, y)	0.64	0.79	0.77

Note: This table reports aggregate measures for EBM and the two SHAP decomposition over 100 simulation runs. The decompositions are compared based on model fit (R^2 importance rankings) and term-level RMSE.

Table 11: Simulation results

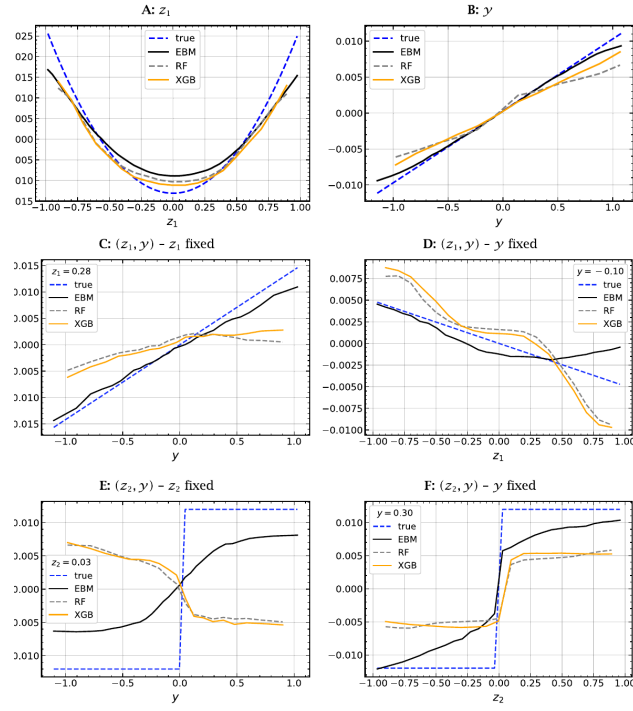


Figure 15: Comparison of fitted effects in simulation

6 Short summary

- The paper studies corporate bond return prediction with an interpretable machine learning model.
- The model is an Explainable Boosting Machine: a generalized additive model with selected nonlinear pairwise interactions.
- The target is next-month firm-level corporate bond excess returns.
- The data cover 81 predictors, 1,207 firms, 106,265 firm-month observations, and July 2002 to August 2020.
- EBM achieves 12.10% out-of-sample R^2 , close to Random Forest and XGBoost and much better than OLS and LASSO.
- The most important predictors are changes in financial uncertainty, changes in macroeconomic uncertainty, TERM, LTREVB, and spread.
- A parsimonious model with ten variables captures almost the full model’s out-of-sample performance.
- Predictor effects are highly nonlinear and often inactive except in the tails.
- TERM predicts high returns mainly when long Treasury returns strongly exceed short Treasury bill returns.
- Rising financial uncertainty predicts lower corporate bond returns.
- Falling financial uncertainty predicts higher returns, especially for high-spread and low-grade bonds.
- Small firms, long-maturity bonds, and high-risk bonds are more sensitive to uncertainty.
- PCA macro factors cannot replicate the predictive role of the uncertainty variables.
- EBM portfolios sorted on predicted returns have strong long-short Sharpe ratios.
- The EBM portfolio’s composition is transparent: high predicted returns are linked to spread and uncertainty interactions.
- SHAP explanations are similar for many univariate effects, but less reliable for interactions.
- Simulations confirm that EBM recovers nonlinear interaction effects more accurately than SHAP decompositions of Random Forest and XGBoost.
- The broad lesson is that interpretability and strong forecasting performance can coexist in asset pricing.

7 Conclusion

7.1 Core conclusion

Bell, Kakhbod, Lettau, and Nazemi show that interpretable machine learning can match black-box predictive performance in corporate bond returns. The EBM reaches state-of-the-art out-of-sample accuracy while preserving a transparent additive structure that shows which predictors and interactions drive each forecast.

7.2 Economic intuition

Corporate bond returns are strongly shaped by uncertainty and credit risk. But the relationship is not linear. Uncertainty matters primarily through tail events, and its effect depends on bond characteristics. When financial uncertainty falls, high-spread bonds experience large positive expected-return contributions, consistent with price recovery and compensation for bearing credit risk. When uncertainty rises, risky bonds are penalized.

7.3 Takeaway

The paper turns machine learning from a forecasting black box into an economic diagnostic tool. The important finding is not only that bond returns are predictable, but that the predictability can be explained by nonlinear uncertainty effects, term structure effects, and interactions between aggregate uncertainty and firm-level bond risk.